Quantum Tomography for High Energy Physics

N. Viaux, A. Norambuena, F. Aguilar-Carvajal

May 2026

1 Quantum Tomography at work

1.1 One Spin-1/2 system

Let's describe the one spin-1/2 density matrix:

$$\rho = \frac{1}{2}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma}\right) \tag{1}$$

If we want to analyze a decay $\tau/\overline{\tau} \to l^+/l^-$, then the option of meassure just two directions (up/down) isn't available, instead we need to meassure in a space, in a solid angle space. So, we have the intention of meassure in a specific direction in which the product of the decay goes on, say $\hat{n}$, then the proyector operator will take the form:

$$\Pi_{\hat{n}} = \frac{1}{2}\left(\mathcal{I} + \hat{n}\cdot\vec{\sigma}\right) \tag{2}$$

The probability that the decay product be emitted in the $\hat{n}$ direction is:

$$\begin{aligned}
\mathrm{p}(\hat{n}) &= \mathrm{Tr}\left[\frac{1}{2}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma}\right)\frac{1}{2}\left(\mathcal{I} + \hat{n}\cdot\vec{\sigma}\right)\right] \\
&= \mathrm{Tr}\left[\frac{1}{4}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma} + \hat{n}\cdot\vec{\sigma} + \underbrace{(\vec{B}\cdot\vec{\sigma})(\hat{n}\cdot\vec{\sigma})}_{(\vec{B}\cdot\hat{n})I + i(\vec{B}\times\hat{n})\cdot\vec{\sigma}}\right)\right] \\
&= \frac{1}{4}\mathrm{Tr}\left[\mathcal{I} + \vec{B}\cdot\vec{\sigma} + \hat{n}\cdot\vec{\sigma} + (\vec{B}\cdot\hat{n})I + i(\vec{B}\times\hat{n})\cdot\vec{\sigma}\right] \\
&= \frac{1}{4}\mathrm{Tr}\left[\mathcal{I} + \left(\vec{B} + \hat{n} + i(\vec{B}\times\hat{n})\right)\cdot\vec{\sigma} + (\vec{B}\cdot\hat{n})\mathcal{I}\right] \\
&= \frac{1}{4}\left[\underbrace{\mathrm{Tr}\left[\mathcal{I}\right]}_{4} + \underbrace{\mathrm{Tr}\left[\left(\vec{B} + \hat{n} + i(\vec{B}\times\hat{n})\right)\cdot\vec{\sigma}\right]}_{\vec{\sigma}\text{'s are traceless}} + \mathrm{Tr}\left[(\vec{B}\cdot\hat{n})I\right]\right] \\
&= \frac{1}{4}\left[2 + 2\vec{B}\cdot\hat{n}\right]
\end{aligned} \tag{3}$$

So, the probability is given by:

$$\mathrm{p}(\hat{n}) = \frac{1}{2}\left[1 + \vec{B}\cdot\hat{n}\right] \tag{4}$$

However, the probability isn't normalized in the solid angle space. Let's say that $\mathrm{P}(\hat{n}) \propto 1 + \vec{B}\cdot\hat{n}$, then:

$$\int d\Omega\ \mathrm{p}(\hat{n}) = 1 \tag{5}$$

$$\int d\Omega\left[c(1 + \vec{B}\cdot\hat{n})\right] = 1 \tag{6}$$

$$c\underbrace{\int d\Omega}_{4\pi} + c\vec{B}\cdot\underbrace{\int d\Omega\ \hat{n}}_{0} = 1 \tag{7}$$

$$c = \frac{1}{4\pi} \tag{8}$$

Finally, the probability density is given by:

$$\mathrm{p}(\hat{n}) = \frac{1}{4\pi}\left[1 + \vec{B}\cdot\hat{n}\right] \tag{9}$$

Finally, to perform Quantum Tomography we need to include a *Spin-analyzing power* parameter, κ, in order to cunatify how efficiently the $\hat{n}$ direction tracks the original polarization of the mother particle $\vec{B}$.

$$\mathrm{p}(\hat{n}) = \frac{1}{4\pi}\left[1 + \kappa\vec{B}\cdot\hat{n}\right] \tag{10}$$

1.2 Important step to further probe

An important step that isn't detailled in the literature, and also I still don't completelly understand it, is that to obtain a full probability density function that the decay product be projected ($\hat{n}$) in a particular direction (+ or -), is multiply the probabilities of each path; that is:

$$\mathrm{p}(\hat{n}^+,\hat{n}^-) = \frac{1}{4\pi}\left[1 + \kappa_+ B_i^+ n_i^+\right]\frac{1}{4\pi}\left[1 + \kappa_- B_j^- n_j^-\right] \tag{11}$$

$$\mathrm{p}(\hat{n}^+,\hat{n}^-) = \frac{1}{(4\pi)^2}\left[1 + \kappa_+ B_i^+ n_i^+ + \kappa_- B_j^- n_j^- + \kappa_+\kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^-\right] \tag{12}$$

1.3 Expected values

Since we already have an expression for the probability density function, the next step is try to find an expected value for a particular direction for the decay product.

1.3.1 $\langle \hat{n}^+ \rangle$

$$
\begin{aligned}
\langle n_i^+ \rangle &= \int d\Omega^+ d\Omega^- n_l^+ \mathrm{p}(n^+, n^-) \\
&= \int d\Omega^+ d\Omega^- n_l^+ \frac{1}{(4\pi)^2} \left[1 + \kappa_+ B_i^+ n_i^+ + \kappa_- B_j^- n_j^- + \kappa_+\kappa_- \sum_{i,j} C_{ij} n_i^+ n_j^- \right] \\
&= \frac{1}{(4\pi)^2} \int d\Omega^+ d\Omega^- \left[n_l^+ + \kappa_+ B_i^+ n_l^+ n_i^+ + \kappa_- B_j^- n_l^+ n_j^- + \kappa_+\kappa_- \sum_{i,j} C_{ij} n_k^+ n_i^+ n_j^- \right] \\
&= \frac{1}{(4\pi)^2} \int d\Omega^+ d\Omega^- \left[\underbrace{n_l^+}_{(1)} + \underbrace{\kappa_+ B_i^+ n_l^+ n_i^+}_{(2)} + \underbrace{\kappa_- B_j^- n_l^+ n_j^-}_{(3)} + \underbrace{\kappa_+\kappa_- \sum_{i,j} C_{ij} n_l^+ n_i^+ n_j^-}_{(4)} \right]
\end{aligned} \tag{13}
$$

$$
(1): \quad \int d\Omega^+ d\Omega^- n_l^+ = \int \underbrace{d\Omega^-}_{4\pi} \int \underbrace{d\Omega^+ n_l^+}_{0} = 0 \tag{14}
$$

$$
(2): \quad \int d\Omega^+ d\Omega^- \kappa_+ B_i^+ n_l^+ n_i^+ = \kappa_+ B_i^+ \int \underbrace{d\Omega^-}_{4\pi} \int \underbrace{d\Omega^+ n_l^+ n_i^+}_{\frac{4\pi}{3}\delta_{li}} = \frac{(4\pi)^2}{3} \kappa_+ B_l^+ \tag{15}
$$

$$
(3): \quad \int d\Omega^+ d\Omega^- \kappa_- B_j^- n_l^+ n_j^- = \kappa_- B_j^- \int \underbrace{d\Omega^+ n_l^+}_{0} \int \underbrace{d\Omega^- n_j^-}_{0} = 0 \tag{16}
$$

$$
(4): \quad \int d\Omega^+ d\Omega^- \kappa_+\kappa_- \sum_{i,j} C_{ij} n_l^+ n_i^+ n_j^- = \kappa_+\kappa_- \sum_{i,j} C_{ij} \int d\Omega^+ n_l^+ n_i^+ \int \underbrace{d\Omega^- n_j^-}_{0} = 0 \tag{17}
$$

So, the expected value for the direction n_i^+ for the decay product is:

$$
\langle n_i^+ \rangle = \frac{\kappa_+ B_i^+}{3} \tag{18}
$$

or, generally:

$$\langle n_i^{\pm} \rangle = \frac{\kappa_{\pm} B_i^{\pm}}{3} \tag{19}$$